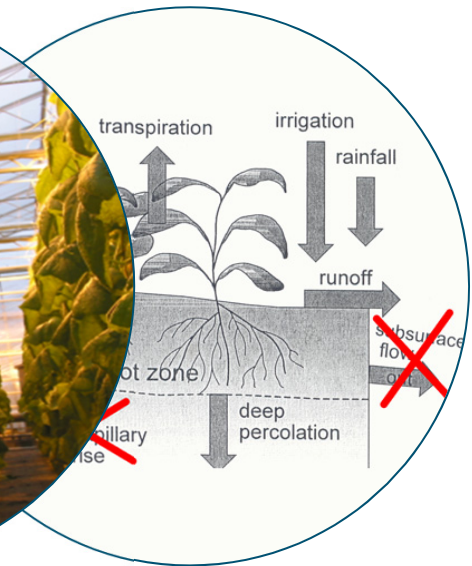
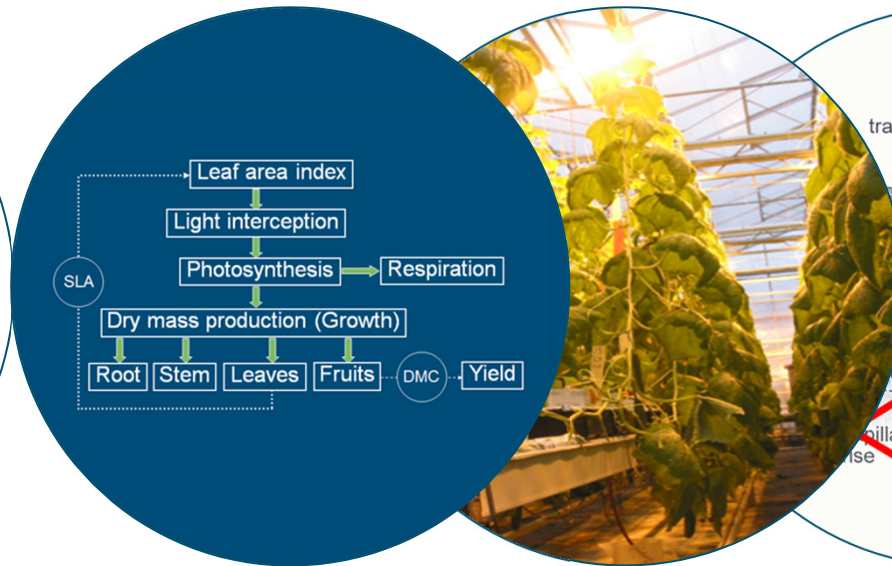
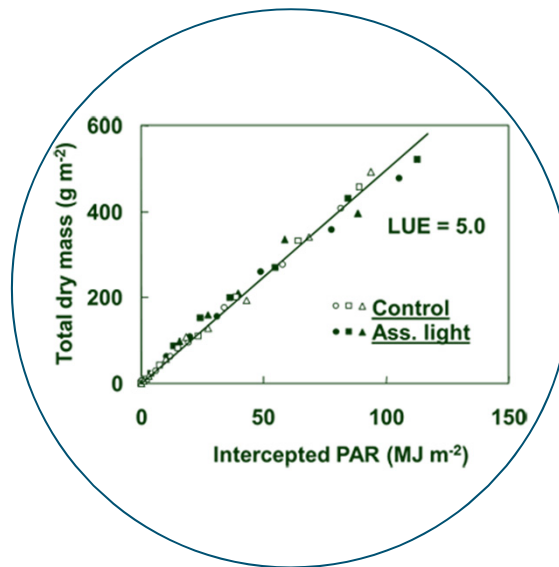


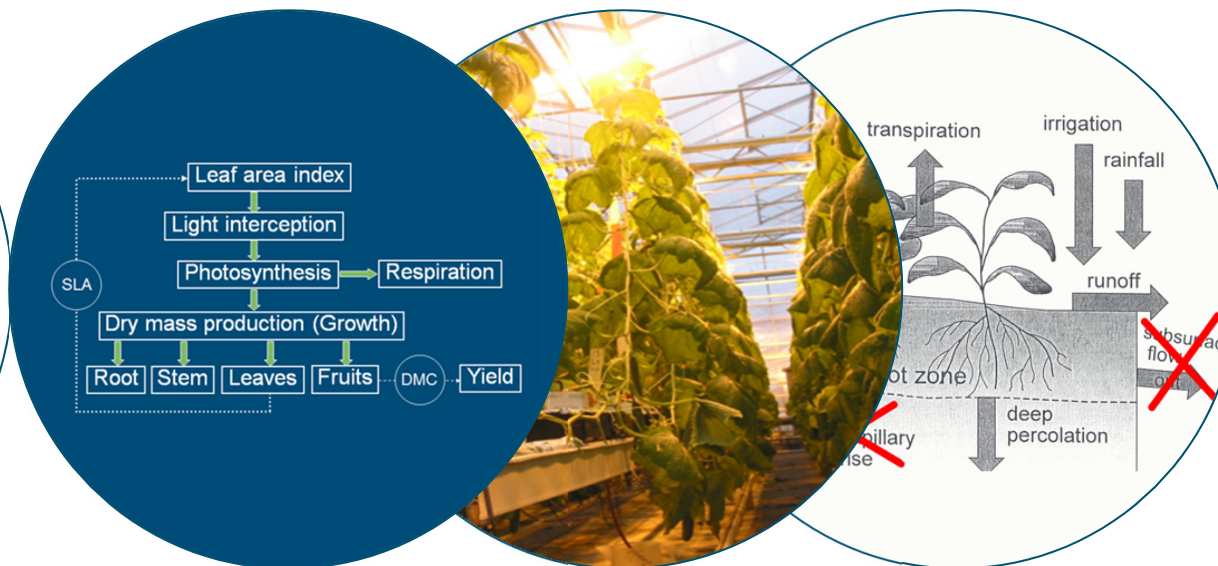
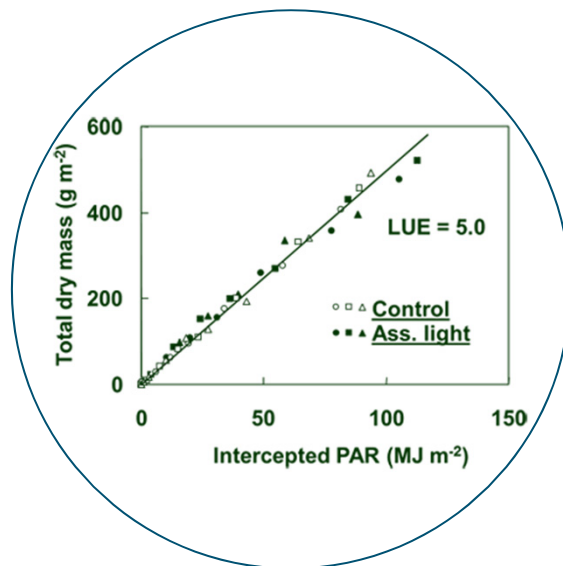
# Lintul1: Potential crop growth

Dr. ir. Peter Leffelaar - Wageningen University



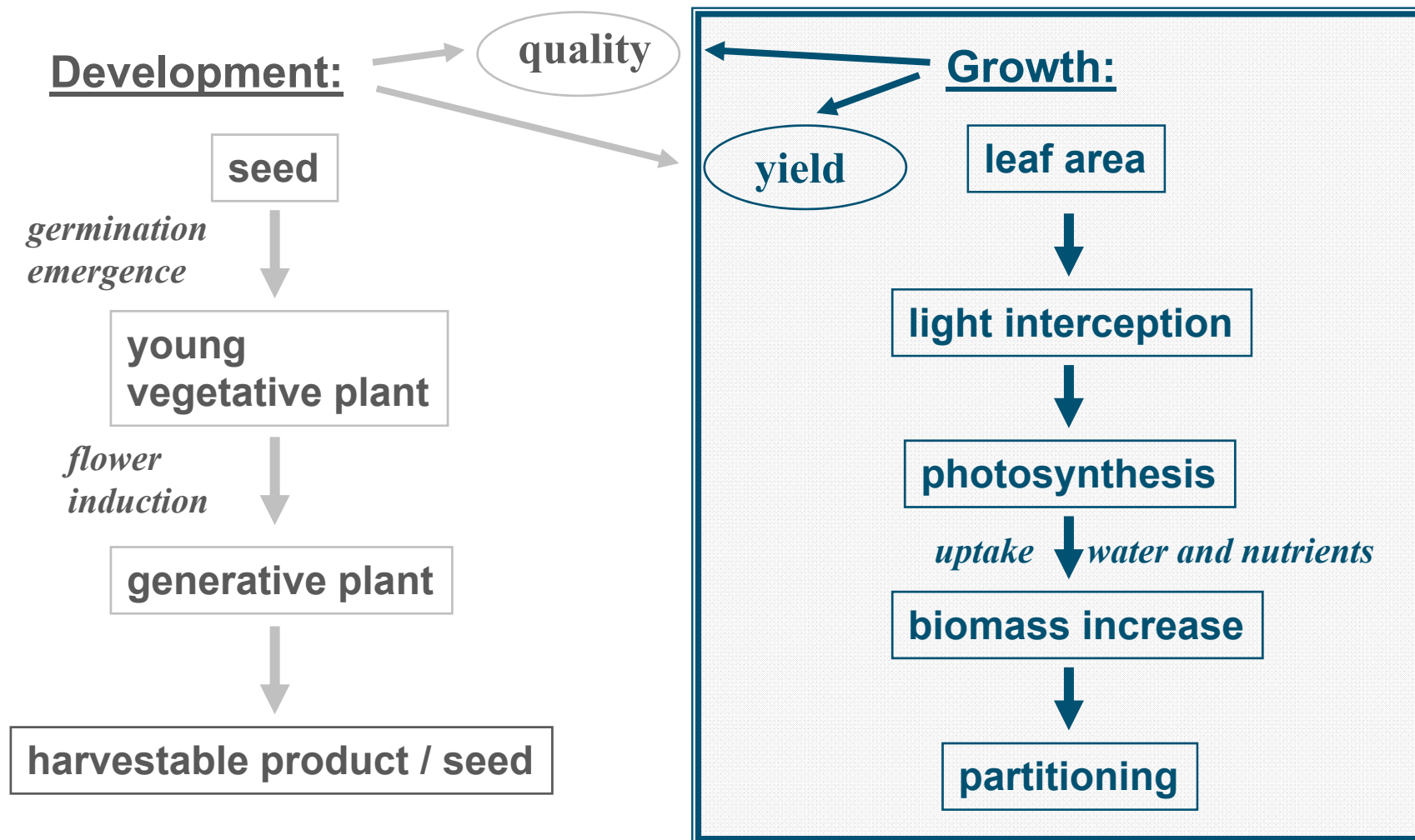
# Lintul1: Potential crop growth

## Introduction

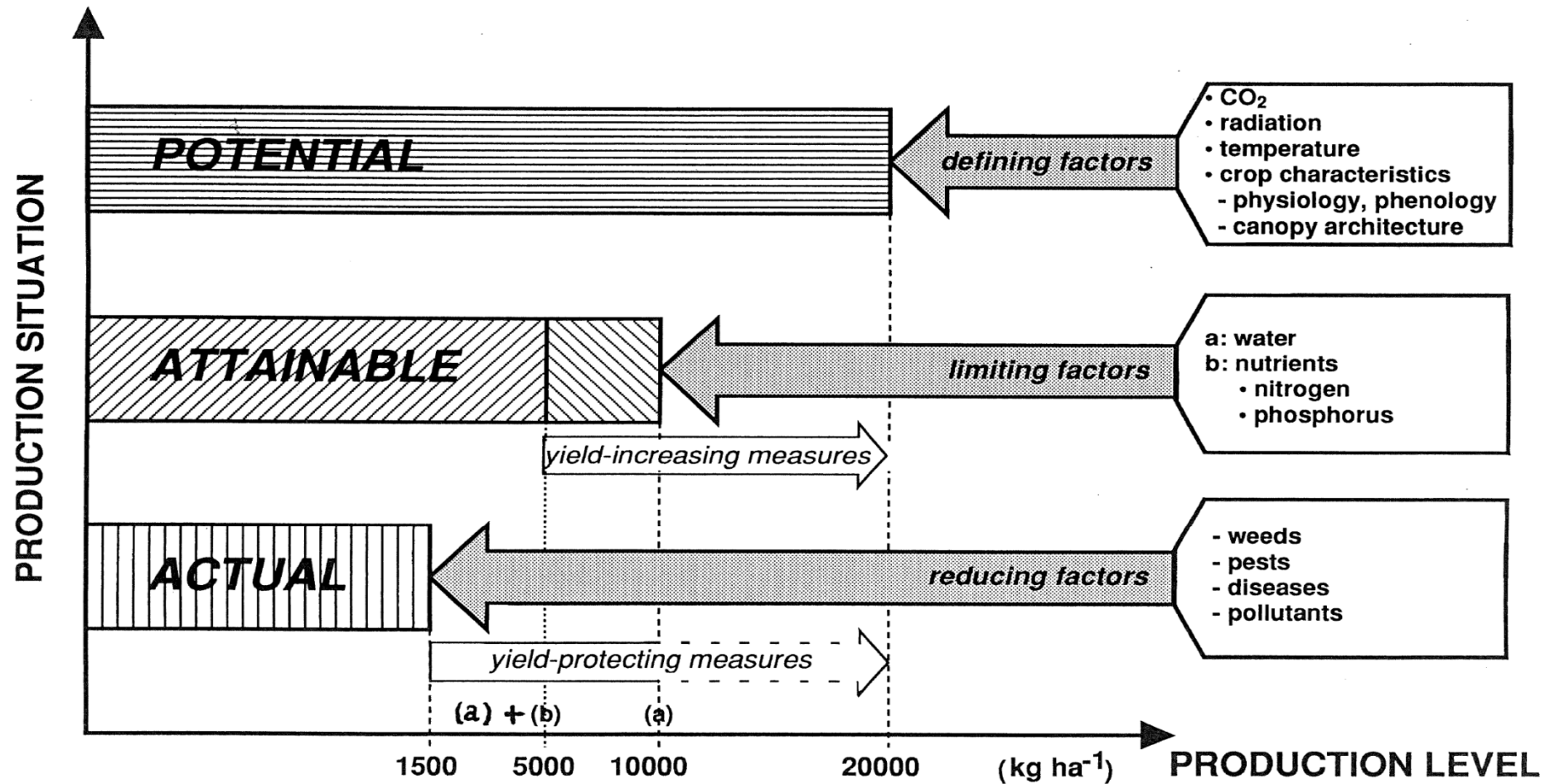


# Lintul1: Potential crop growth

## - scope of the course -



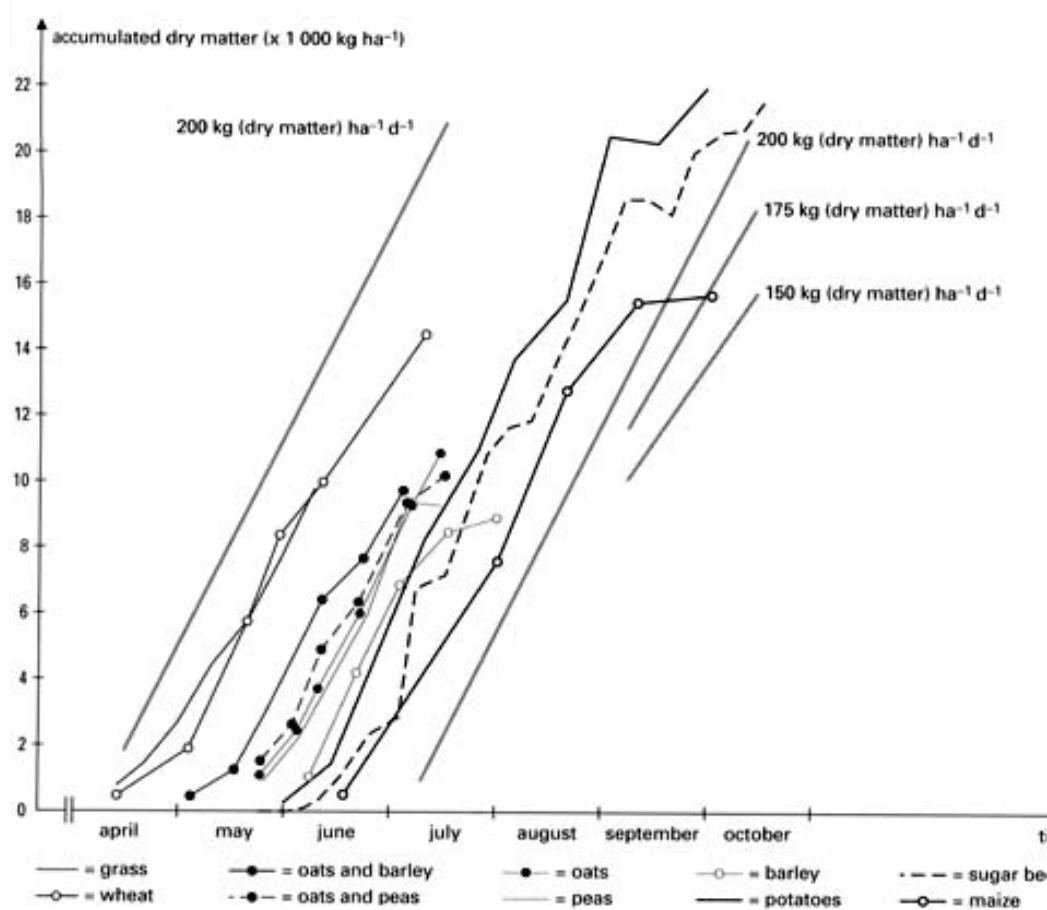
# Potential, attainable and actual production levels



Schematic overview of production situations with corresponding dry matter production levels



# Optimal growth rates - rule of thumb



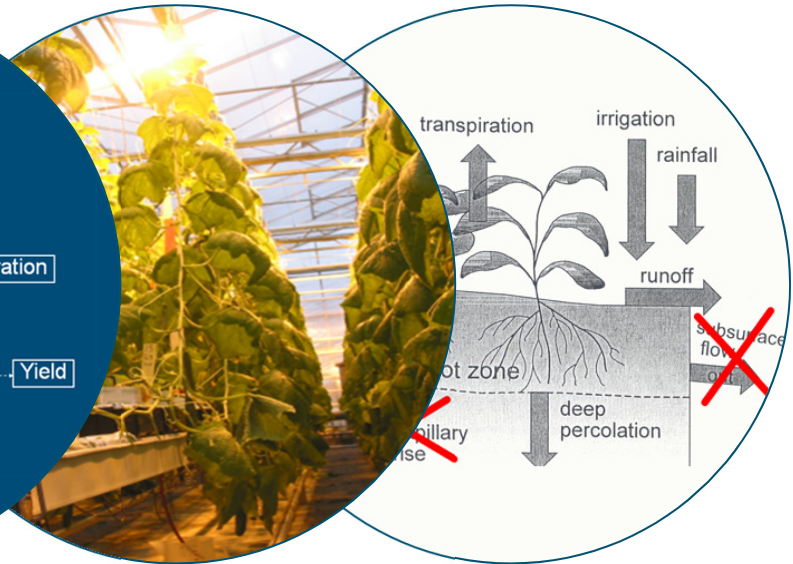
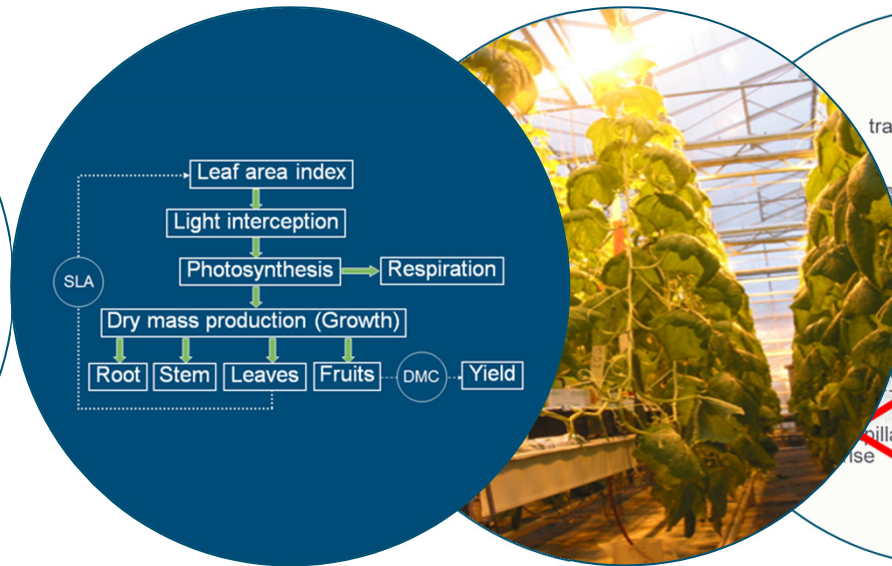
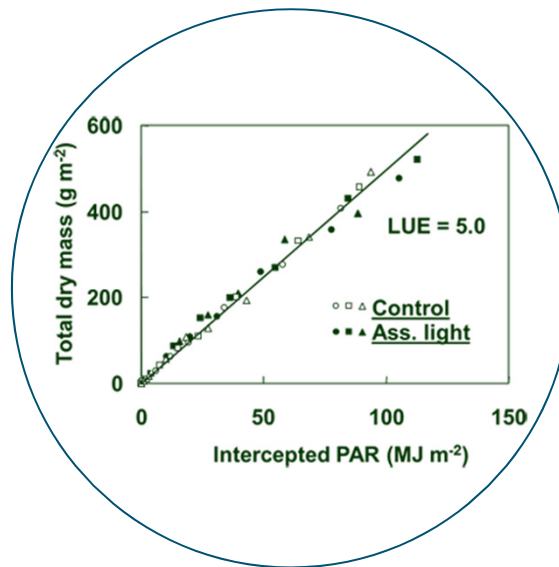
Growth rates of the main agricultural crops in the Netherlands under (near)-optimal growth conditions, compared to growth rates of 200, 175 and 150 kg (dry matter) ha<sup>-1</sup> d<sup>-1</sup>  
(Source: Sibma, 1968)

From: Sibma, 1968

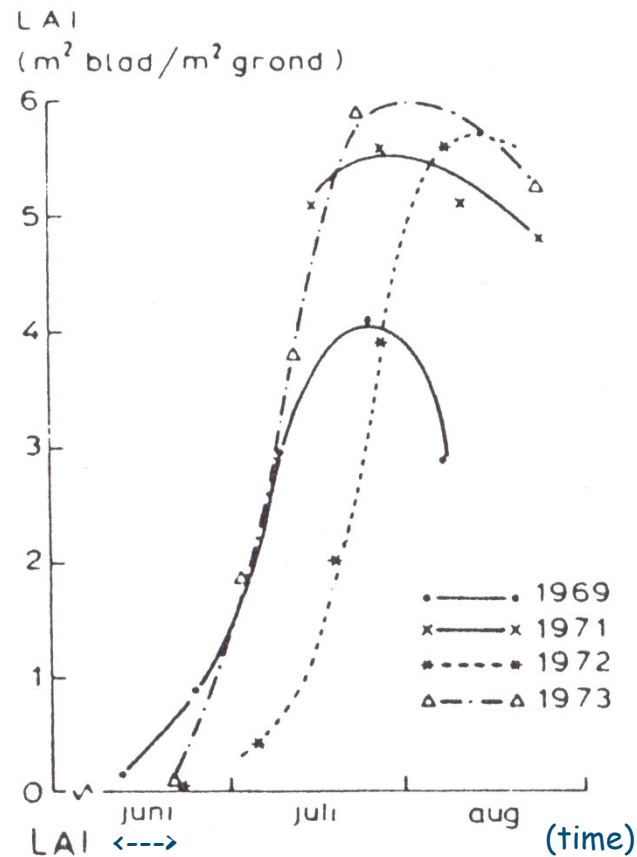


# Lintul1: Potential crop growth

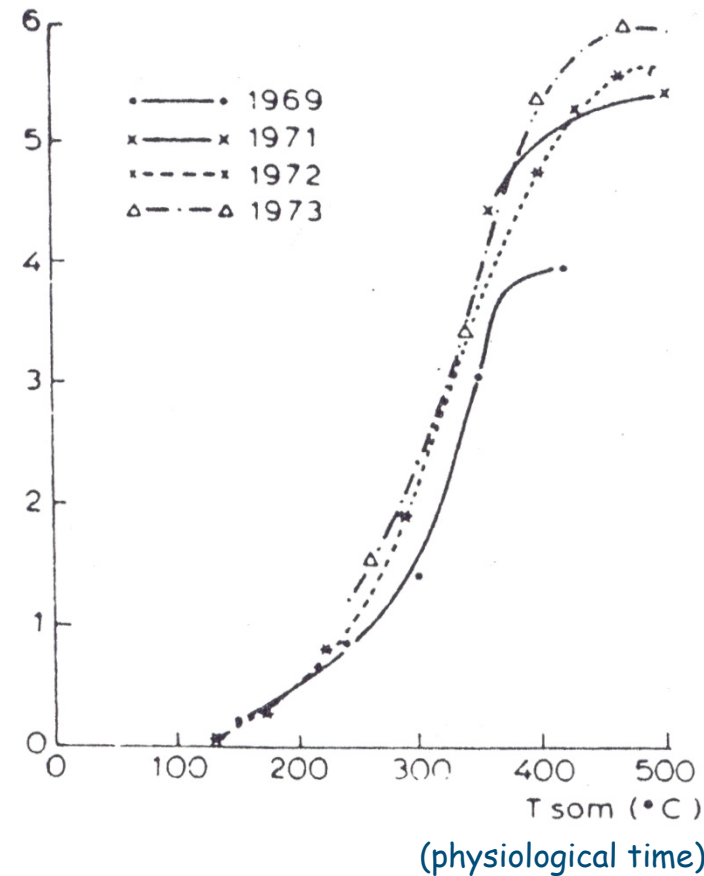
## Light and light interception



# Leaf area of maize as a function of (physiological) time



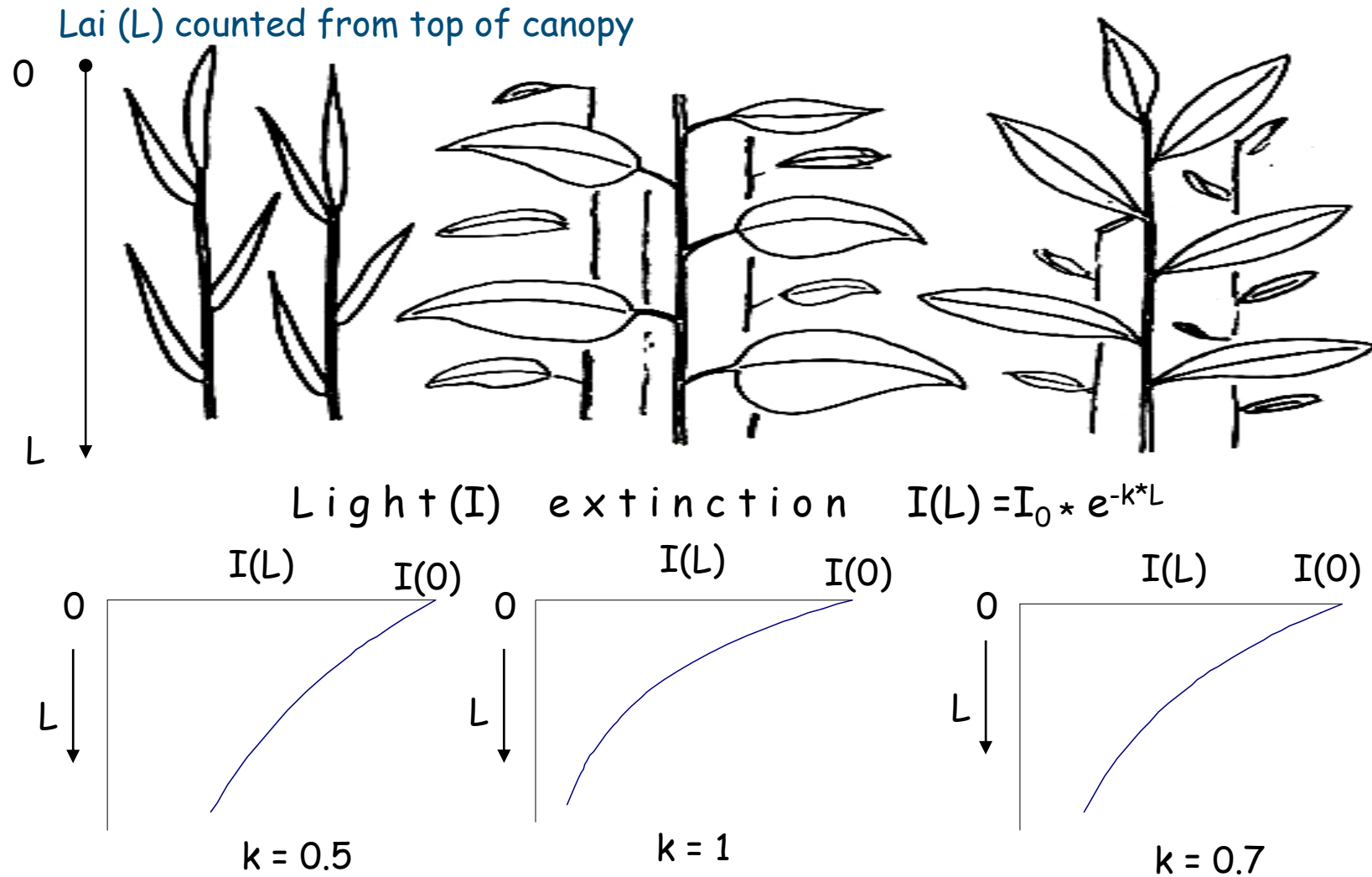
small plants  $\rightarrow$  low LAI  $\rightarrow$   
 not all sunlight used  $\rightarrow$   
 growth rate  $\ll 200 \text{ kg}/(\text{ha} \cdot \text{d})$



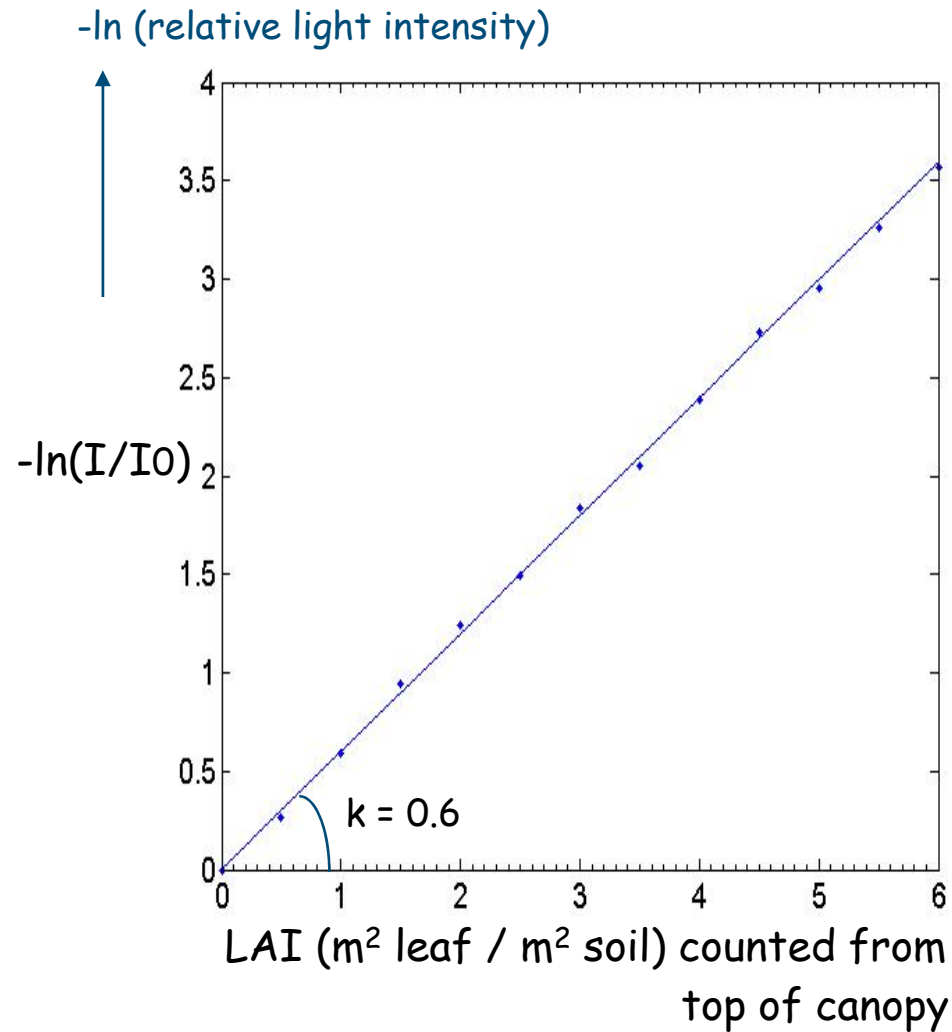
From: Sibma, 1987



# Light extinction is a function of LAI



# Light extinction coefficient, $k$ , follows from slope



$$I(L) = I_0 \cdot e^{-k \cdot L}$$

$$\Rightarrow -\ln\left(\frac{I}{I_0}\right) = k \cdot L$$

**maize**

From: Sibma, 1987



# Spectral composition of global radiation

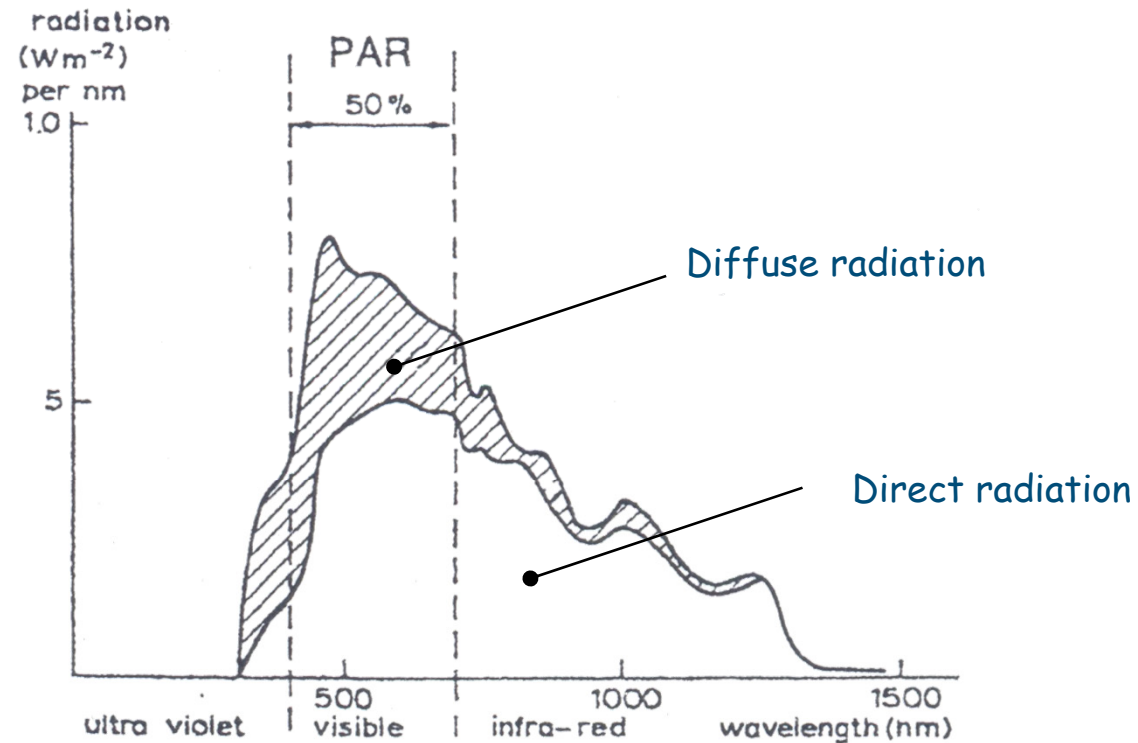


figure 2.2 Spectral composition of global radiation, discriminating between direct radiation (clear area) and diffuse (i.e. indirect) radiation scattered by clouds (hatched area).

PAR = photosynthetically active radiation.

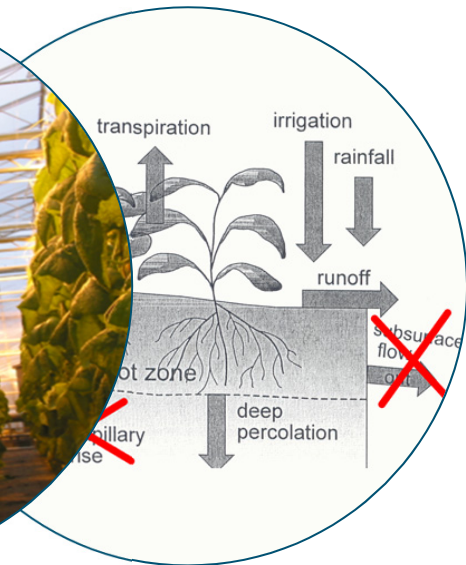
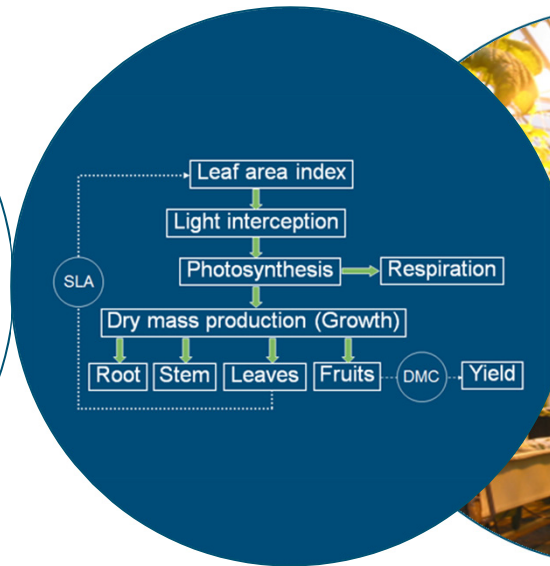
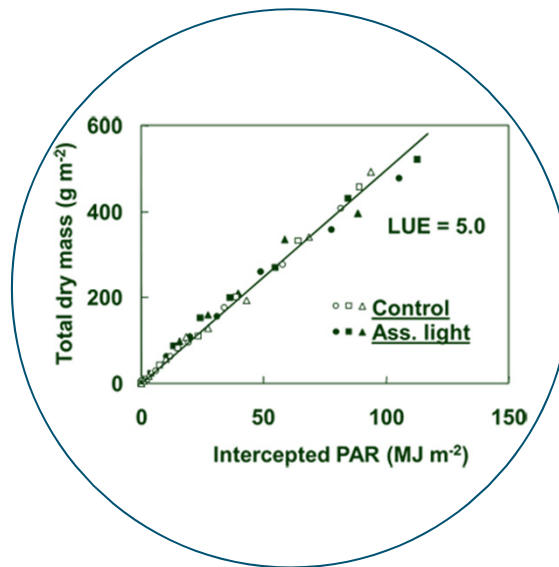
(Source: Monteith and Unsworth, 1990. Principles of environmental physics. Edward Arnold, London, 291 p.)

From: Monteith & Unsworth, 1990

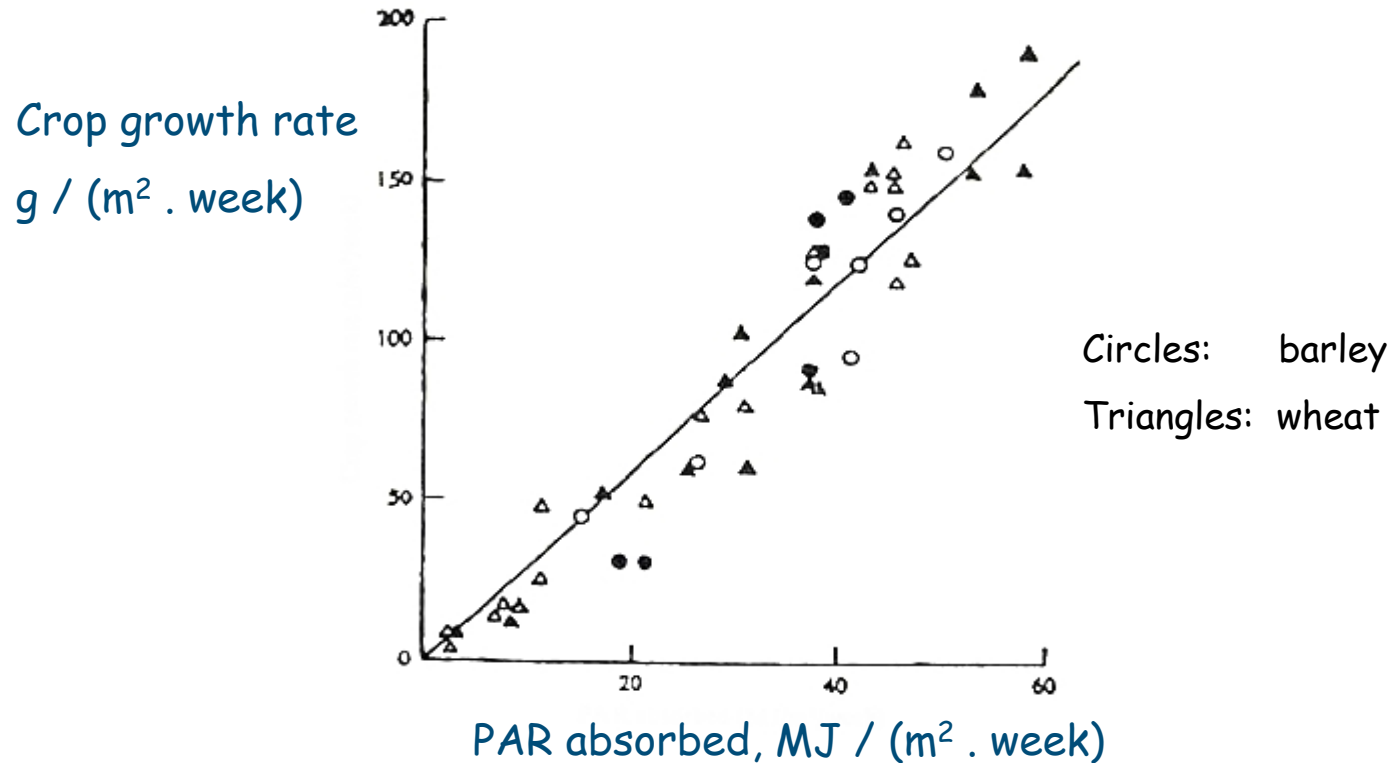


# Lintul1: Potential crop growth

## Light (or Radiation) use efficiency



# Light Use Efficiency, LUE, follows from slope



PARAM LUE = 3.0 (g Dry Matter per MJ PAR)

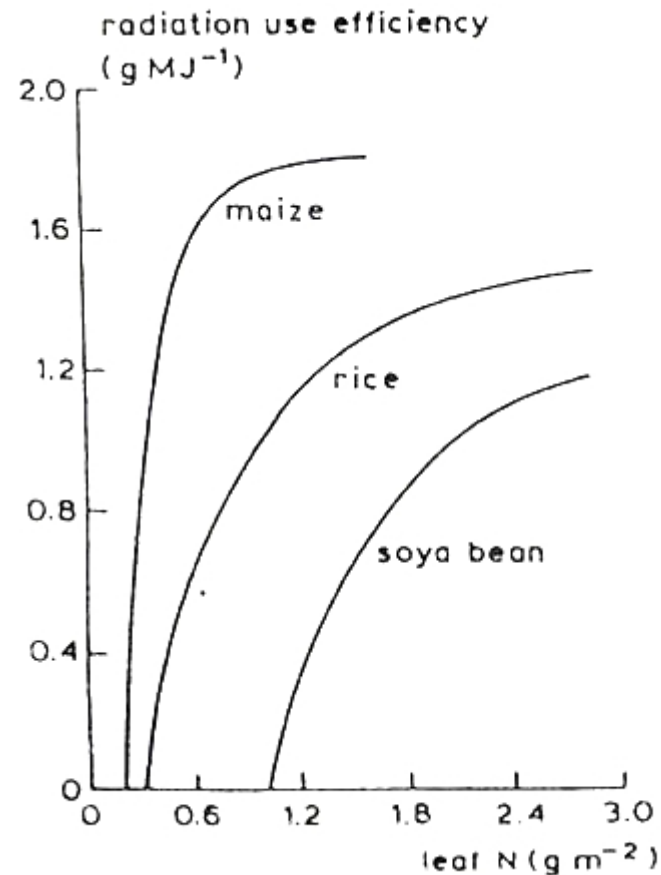
LUE Pinus = 1.9

LUE Quercus = 2.1

From: Gallagher & Biscoe, 1978

# Radiation use efficiency as a function of leaf nitrogen content per unit area

---

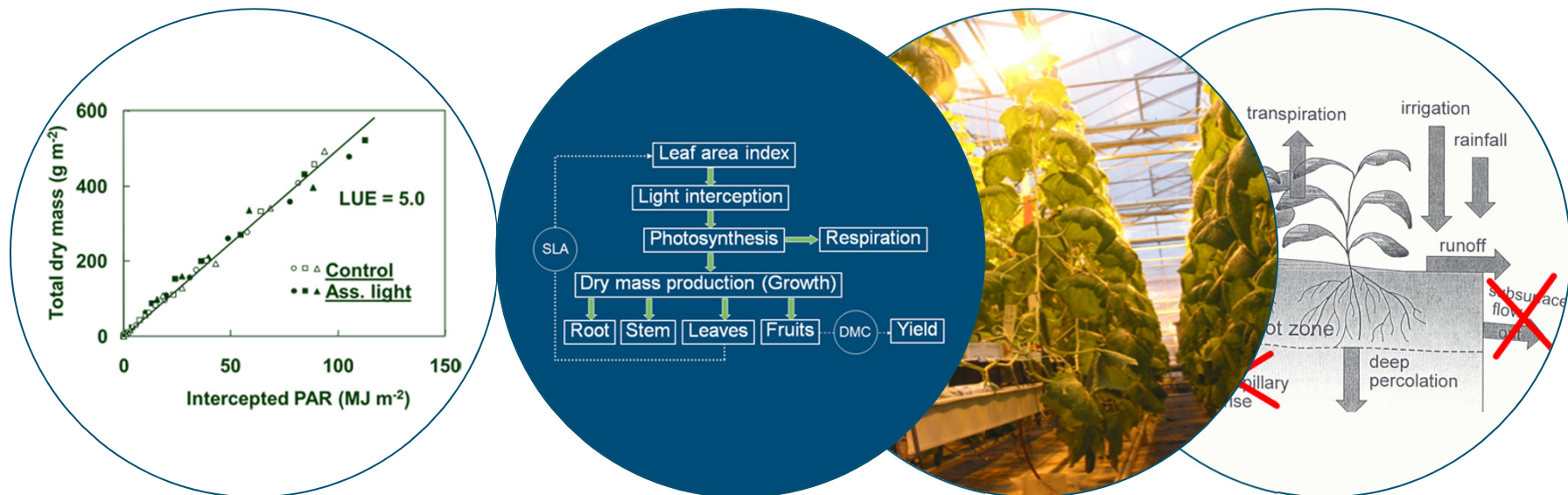


From: Sinclair & Horie, 1989



# Lintul1: Potential crop growth

## Light interception, utilization and growth



# Light interception, utilization and growth in LINTUL

---

TITLE LINTUL

INITIAL

INCON WTOTI = 5.

TIMER STTIME=1. ; FINTIM=100. ; DELT=1. ; PRDEL = 5.

TRANSLATION\_GENERAL DRIVER = 'EUDRIV'

PRINT LAI, WTHA, PARINT

DYNAMIC

LAI = WTOTAL \* LWR \* SLA  
 PARINT = (0.5 \* DTR) \* (1. - EXP(-K \* LAI))

$L = W F_{lv} S_{la}$  compare Eq.(13)  
 $I_{int} = 0.5 DTR (1 - e^{-KL})$  Eq.(2)

GTOTAL = LUE \* PARINT  
 WTOTAL = INTGRL(WTOTI, GTOTAL)  
 WTHA = WTOTAL / 100.

$dW/dt = RUE I_{int}$  Eq.(3)  
 $W = \int dW/dt dt$  Eq.(4)

PARAM DTR = 20. ; LWR = 0.5 ; SLA = 0.022 ; LUE = 3.0 ; K = 0.6

END

STOP

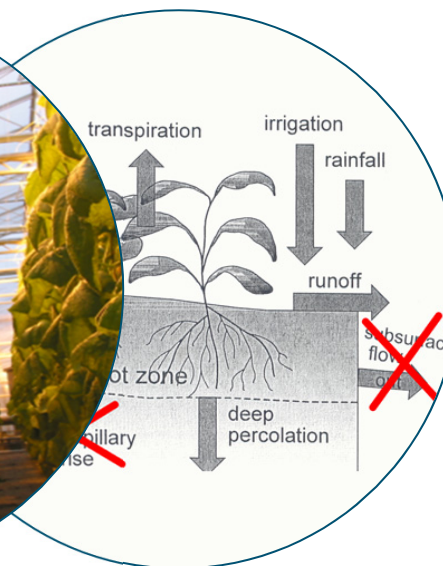
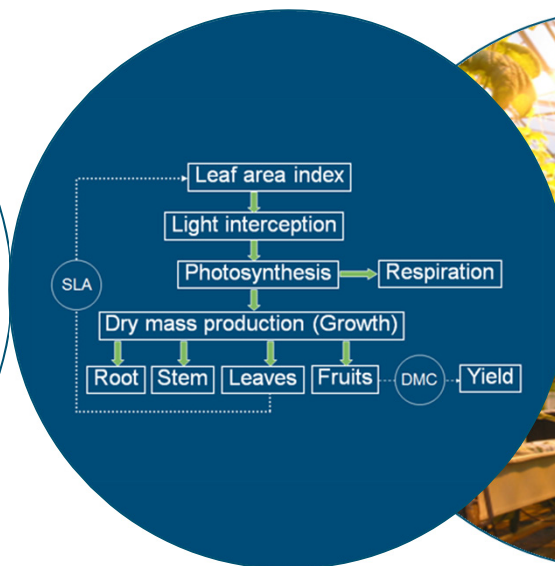
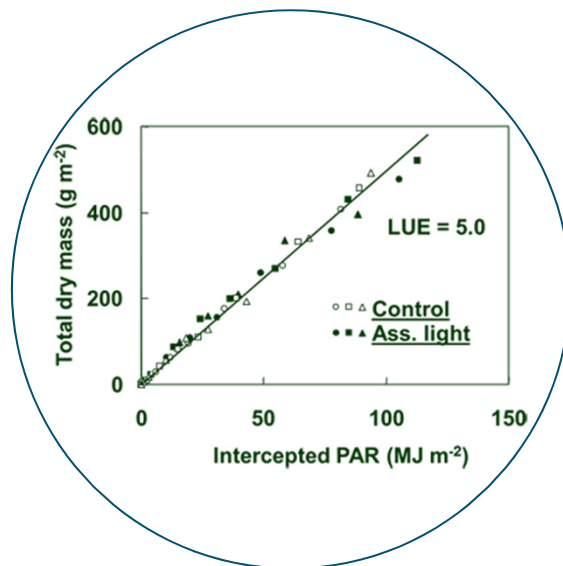
ENDJOB

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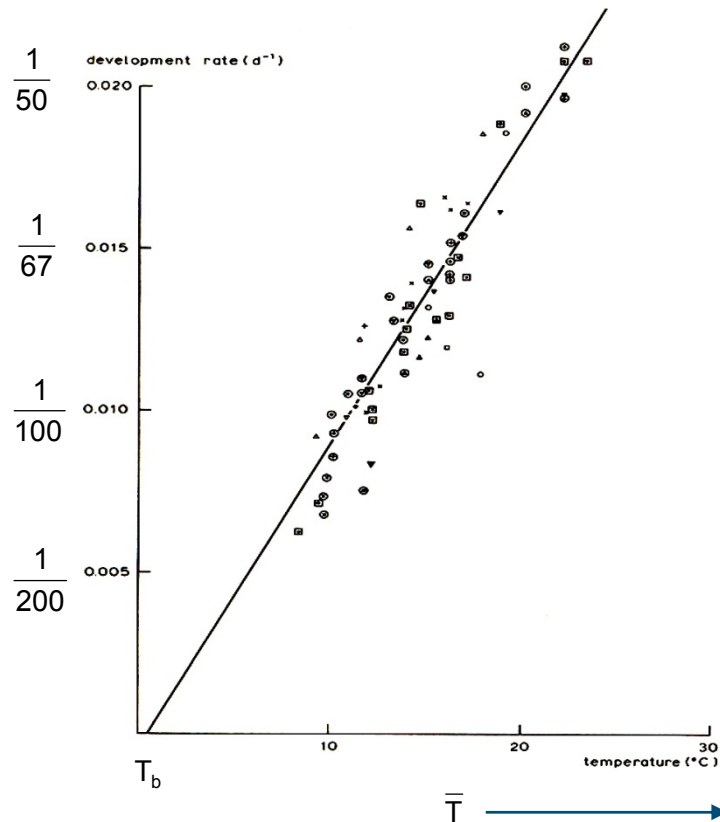
# Lintul1: Potential crop growth

## Crop development



# Development rate of spring wheat as a function of temperature from emergence till flowering . . .

---



$$\frac{1}{\# \text{ days}} = \bar{T} \cdot 0.0009 \Rightarrow$$

$$\text{"temperature sum"} = \bar{T} \cdot \# \text{ days} \Rightarrow$$

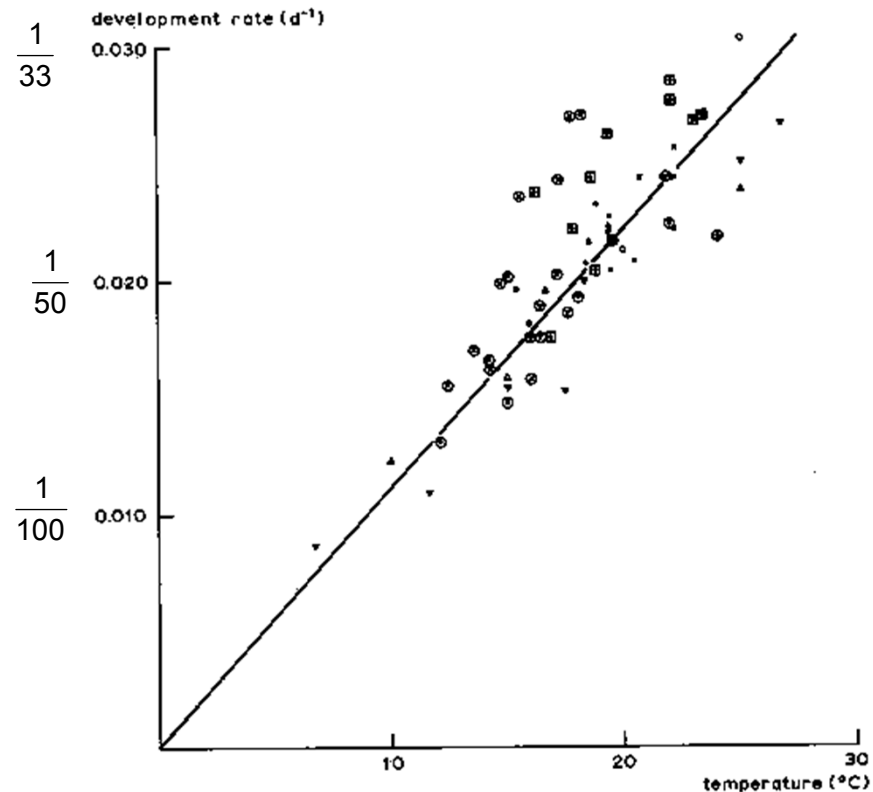
$$\frac{1}{0.0009} = 1110 \text{ }^{\circ}\text{Cd}$$

Data from over 20 papers assembled  
by: Van Keulen & Seligman, 1987



. . . and from flowering to ripeness

---



$$\frac{1}{\# \text{ days}} = \bar{T} \cdot 0.00103 \Rightarrow$$

$$\bar{T} \cdot \# \text{ days} = \frac{1}{0.00103} = 970 \text{ }^{\circ}\text{Cd}$$

Vegetative Phase: 1110 °Cd

Generative Phase: 970 °Cd

Whole growing season: 2080 °Cd

finish condition Lintul

Data from over 20 papers assembled  
by: Van Keulen & Seligman, 1987



# Use of weather data in LINTUL

---

TITLE LINTUL

\* 1. Initial conditions and run control

FINISH **TSUM > 2080.**

\* 2. Environmental data and temperature sum

WEATHER WTRDIR='C:\SYS\WEATHER\'; CNTR='IT'; ISTN=3; ...  
IYEAR=1985

DTR = RDD / 1.E+6

**DAVTMP = 0.5 \* ( TMMN + TMMX )**

**DTEFF = MAX( 0. , DAVTMP - TBASE )**

EMERG = INSW( TIME-DOYEM , 0. , 1. )

**TSUM = INTGRL (ZERO, RTSUM)**

**RTSUM = DTEFF \* EMERG**

$$T_{sum} = \int ( T_a - T_b ) dt$$

\* 6. Functions and parameters for spring wheat

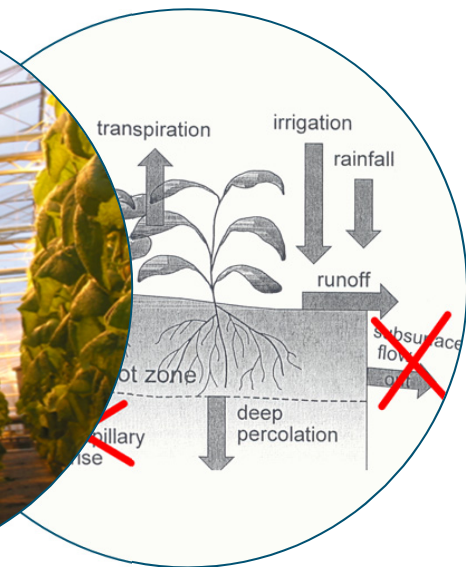
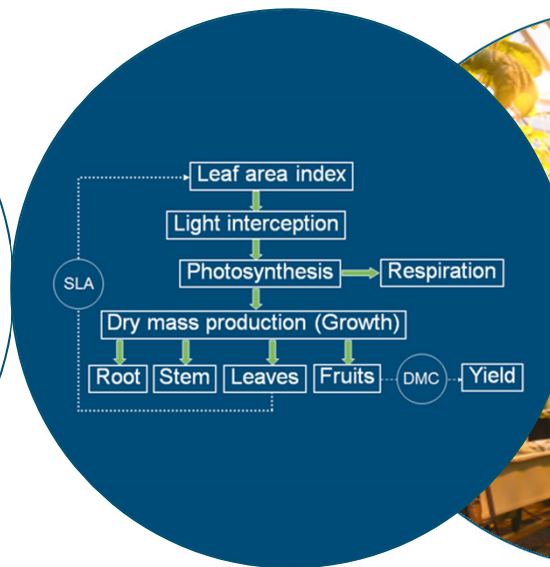
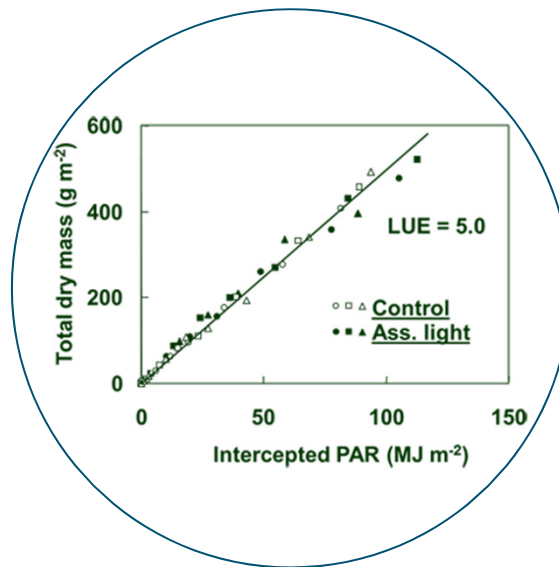
PARAM **TBASE = 0.**

PARAM DOYEM = 32.

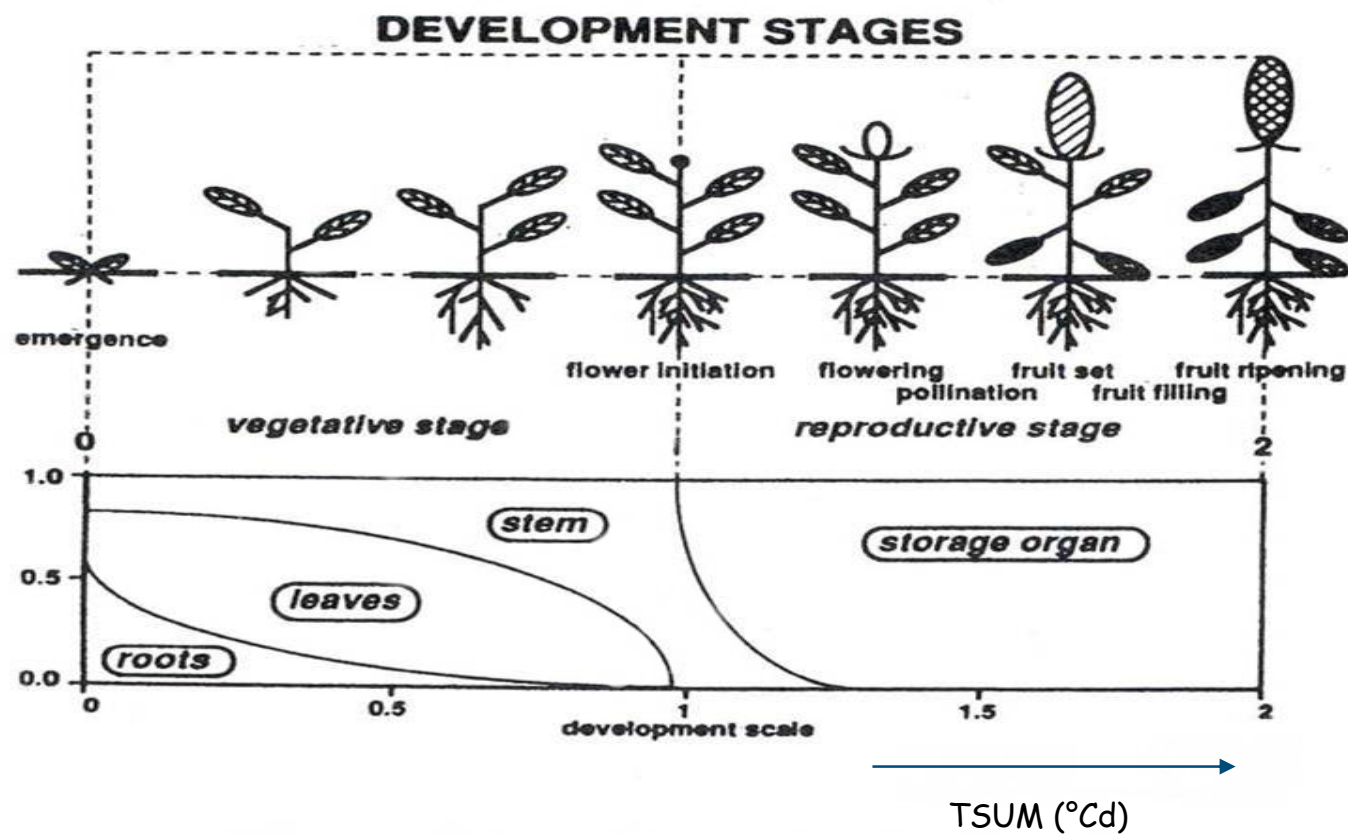


# Lintul1: Potential crop growth

## Development stages and allocation of dry matter



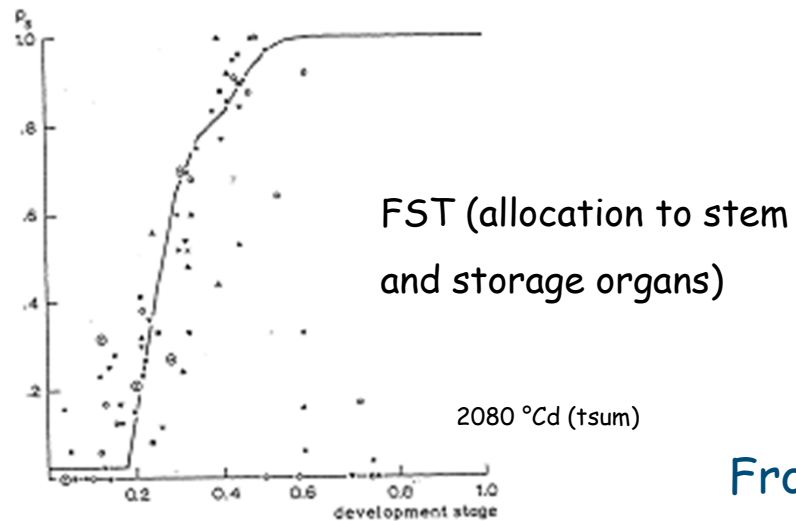
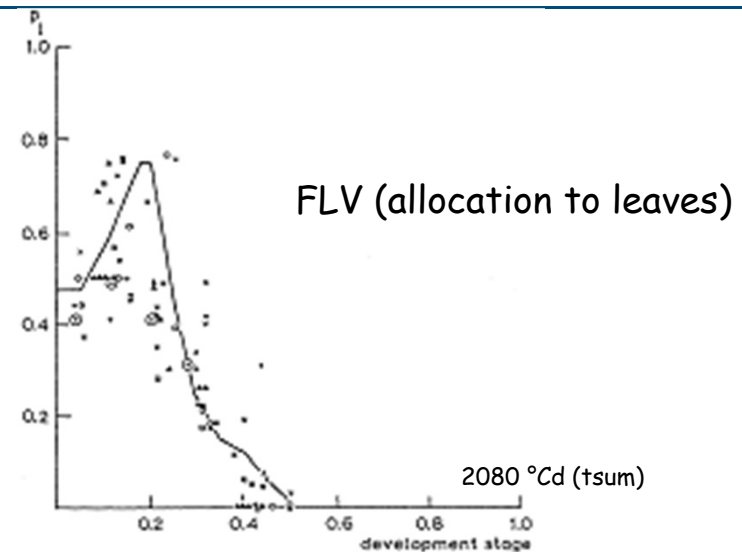
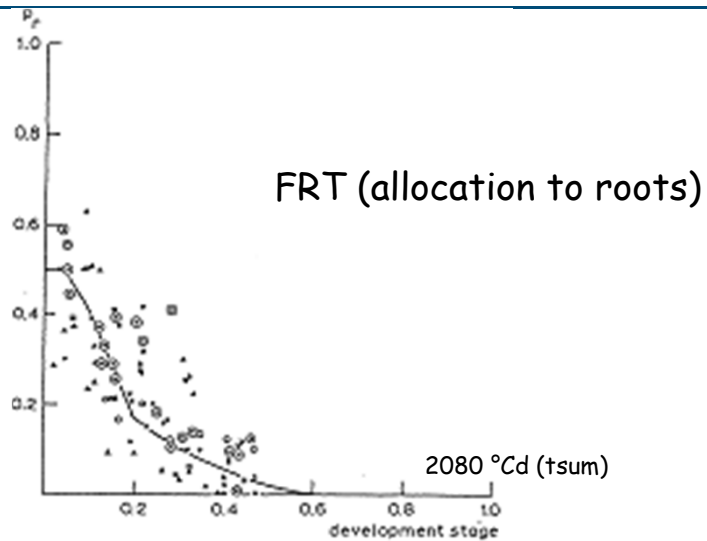
# Crop development stages



From: Lönvenstein, Lantinga, Rabbinge and van Keulen, 1995



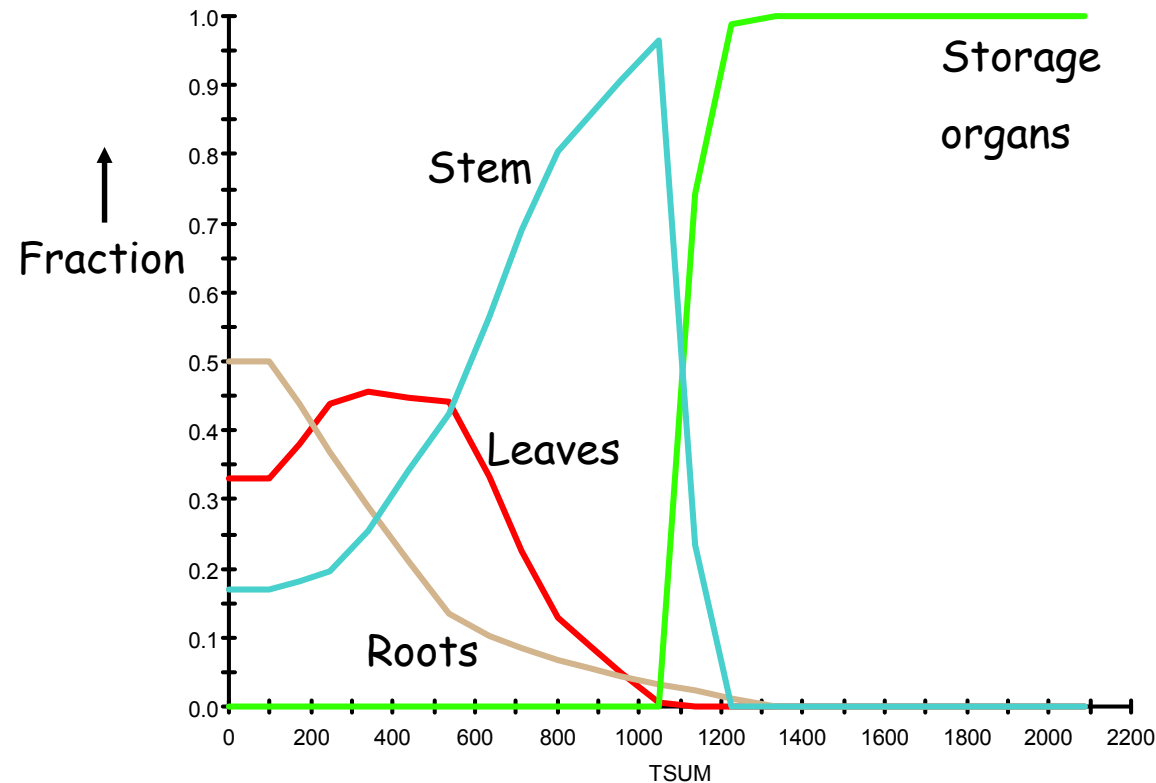
# Substrate allocation patterns in Spring Wheat: Measured



From: van Keulen & Seligman, 1987



## Substrate allocation patterns in Spring Wheat: In model



Allocation pattern of assimilates for spring wheat  
as used in Lintul (Light interception and utilization)



# Growth rates and dry matter production of plant organs

---

TITLE LINTUL

FRT = AFGEN( FRTTB , TSUM )  
FLV = AFGEN( FLVTB , TSUM )  
FST = AFGEN( FSTTB , TSUM )  
FSO = AFGEN( FSOTB , TSUM )

WLVG = INTGRL(WLVI , RWLVG)  
WLVD = INTGRL(ZERO , DLV )  
WST = INTGRL(ZERO , RWST )  
WSO = INTGRL(ZERO , RWSO )

WSOTHA = WSO / 100.

WRT = INTGRL(ZERO , RWRT )  
WLV = WLVG + WLVD

RWLVG = GTOTAL \* FLV - DLV  
RWST = GTOTAL \* FST  
RWSO = GTOTAL \* FSO  
RWRT = GTOTAL \* FRT

$$dW_{lv} / dt = RUE I_{int} F_{lv} - r_d W_{lv} \quad \text{Eq.(8)}$$

$$dW_{st} / dt = RUE I_{int} F_{st} \quad \text{Eq.(7)}$$

$$dW_{so} / dt = RUE I_{int} F_{so} \quad \text{Eq.(7)}$$

$$dW_{rt} / dt = RUE I_{int} F_{rt} \quad \text{Eq.(7)}$$



# Functions for spring wheat

---

TITLE LINTUL

FUNCTION FRTTB = 0.,0.50 , 110., 0.50 , ...  
275.,0.34 , 555.,0.12 , 780.,0.07 , ...  
1055.,0.03 , 1160.,0.02 , 1305.,0. , 2500.,0.

FUNCTION FLVTB = 0.,0.33 , 110.,0.33 , ...  
275.,0.46 , 555.,0.44 , 780.,0.14 , ...  
1055.,0. , 2500.,0.

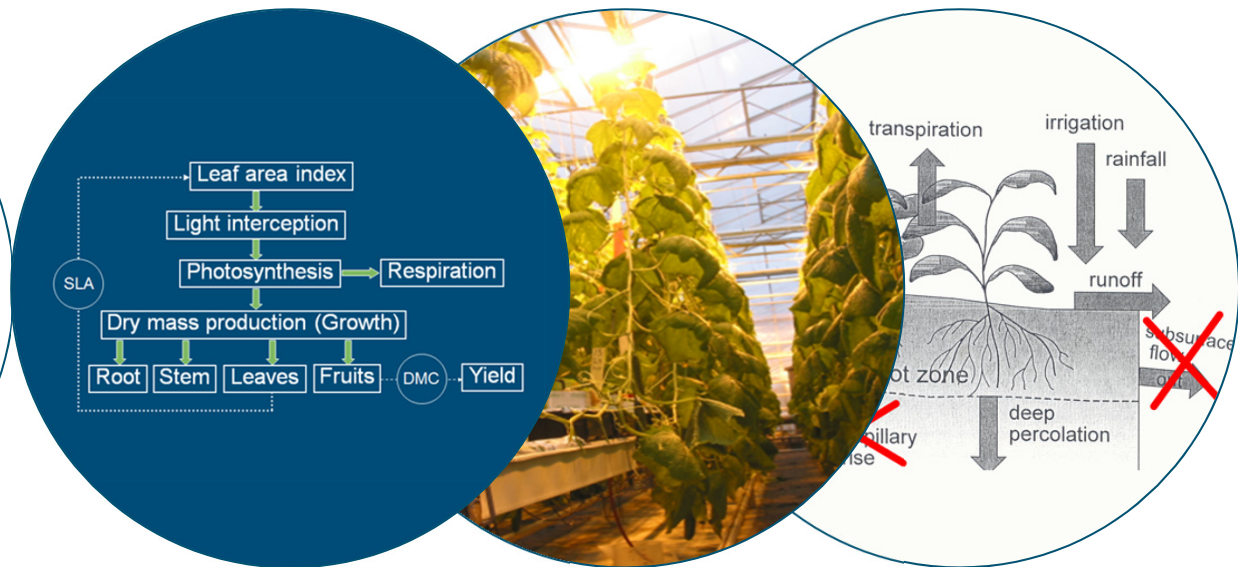
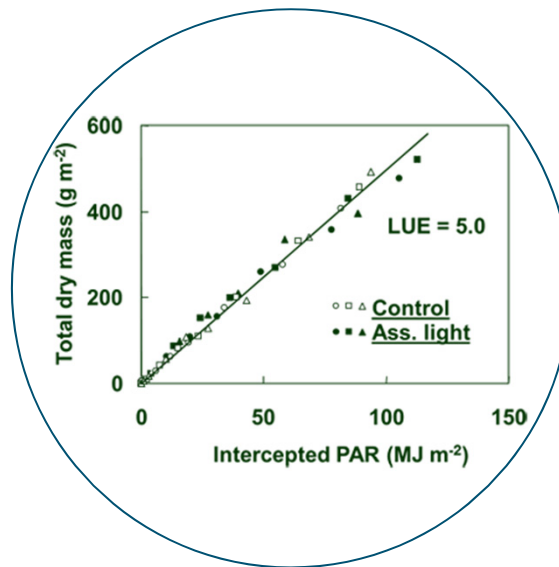
FUNCTION FSTTB = 0.,0.17 , 110.,0.17 , ...  
275., 0.20 , 555., 0.44 , 780., 0.79 , ...  
1055., 0.97 , 1160., 0. , 2500., 0.

FUNCTION FSOTB = 0., 0. , ...  
1055.,0. , 1160.,0.98 , 1305.,1. , 2500.,1.



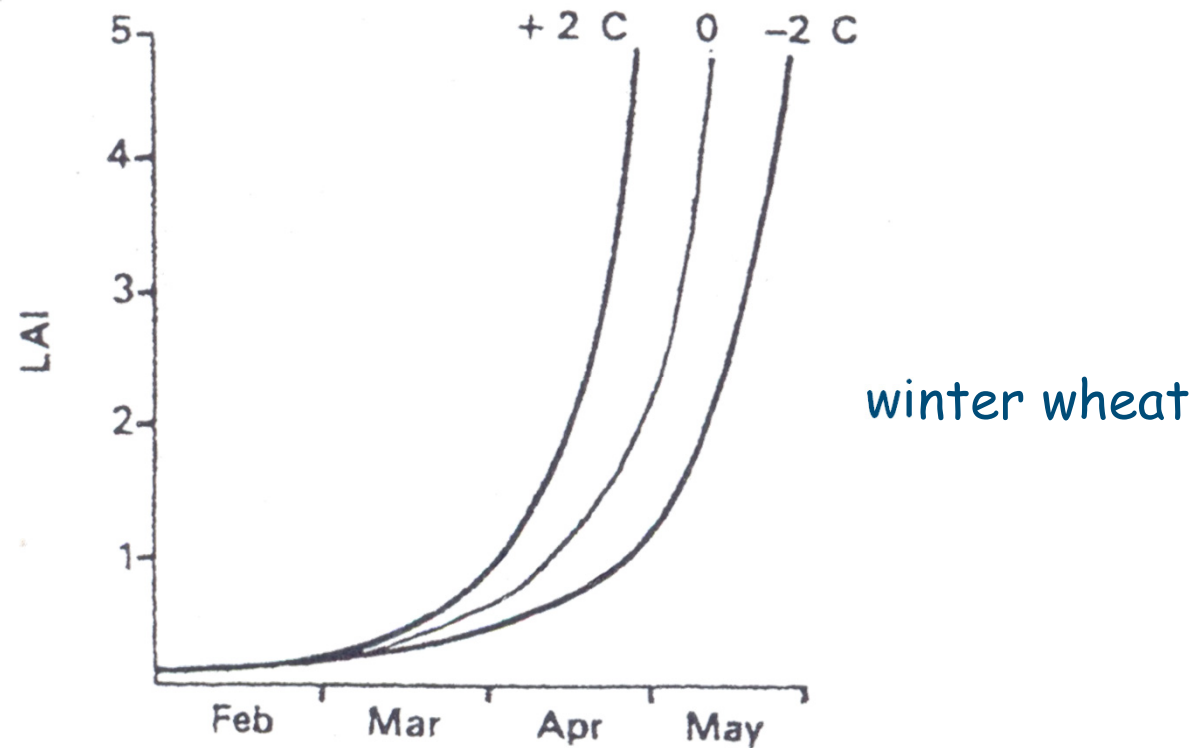
# Lintul1: Potential crop growth

## Early leaf growth



## Early leaf growth: temperature dependent

---



From: Monteith, 1966



# Parameters for spring wheat & Growth of Leaf Area (GLA)

```
PARAM LAII    = 0.012 ; SLA      = 0.022 ; RGRL    = 0.009 ; ...
      TSUMAN = 1110. ; LAICR    = 4.      ; RDRSHM = 0.03
FUNCTION RDRT = -10.,0.03 , 10.,0.03 , 15.,0.04 , 30.,0.09 , 50.,0.09
```

\*-----\*

```
      SUBROUTINE GLA(TIME,DOYEM,DTEFF,TSUM,LAII,RGRL,DELT,
$              SLA ,LAI,GLV,              GLAI)
      IMPLICIT REAL (A-Z)
```

\*---- Growth during **maturation** stage:

**GLAI = SLA \* GLV**

$$dL/dt = S_{la} F_{lv} RUE I_{int}$$

Eq.(7 & 13)

\*---- Growth during **juvenile** stage:

IF ( (TSUM.LT.330.) .AND. (LAI.LT.0.75) )

\$ **GLAI = LAI \* (EXP(RGRL \* DTEFF \* DELT) - 1.) / DELT**

Eq.(12)

\*---- Growth at day of seedling emergence:

IF ( (TIME.GE.DOYEM) .AND. (LAI.EQ.0.) )

\$ GLAI = LAII / DELT

\*---- Growth before seedling emergence:

IF (TIME.LT.DOYEM) GLAI = 0.

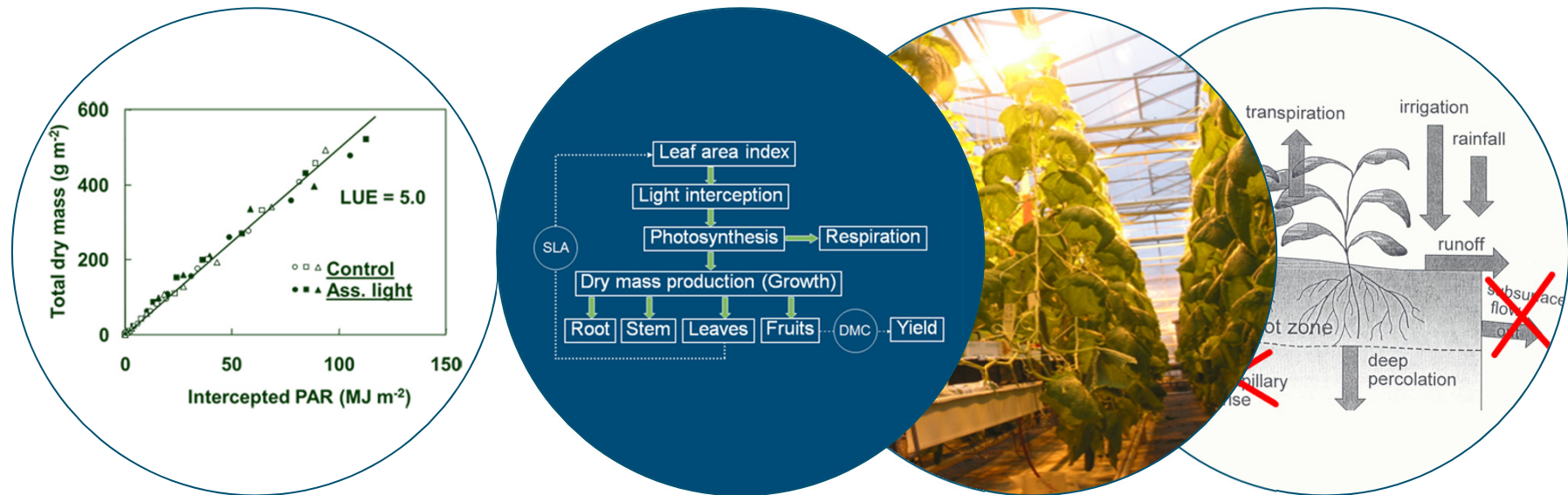
RETURN

END

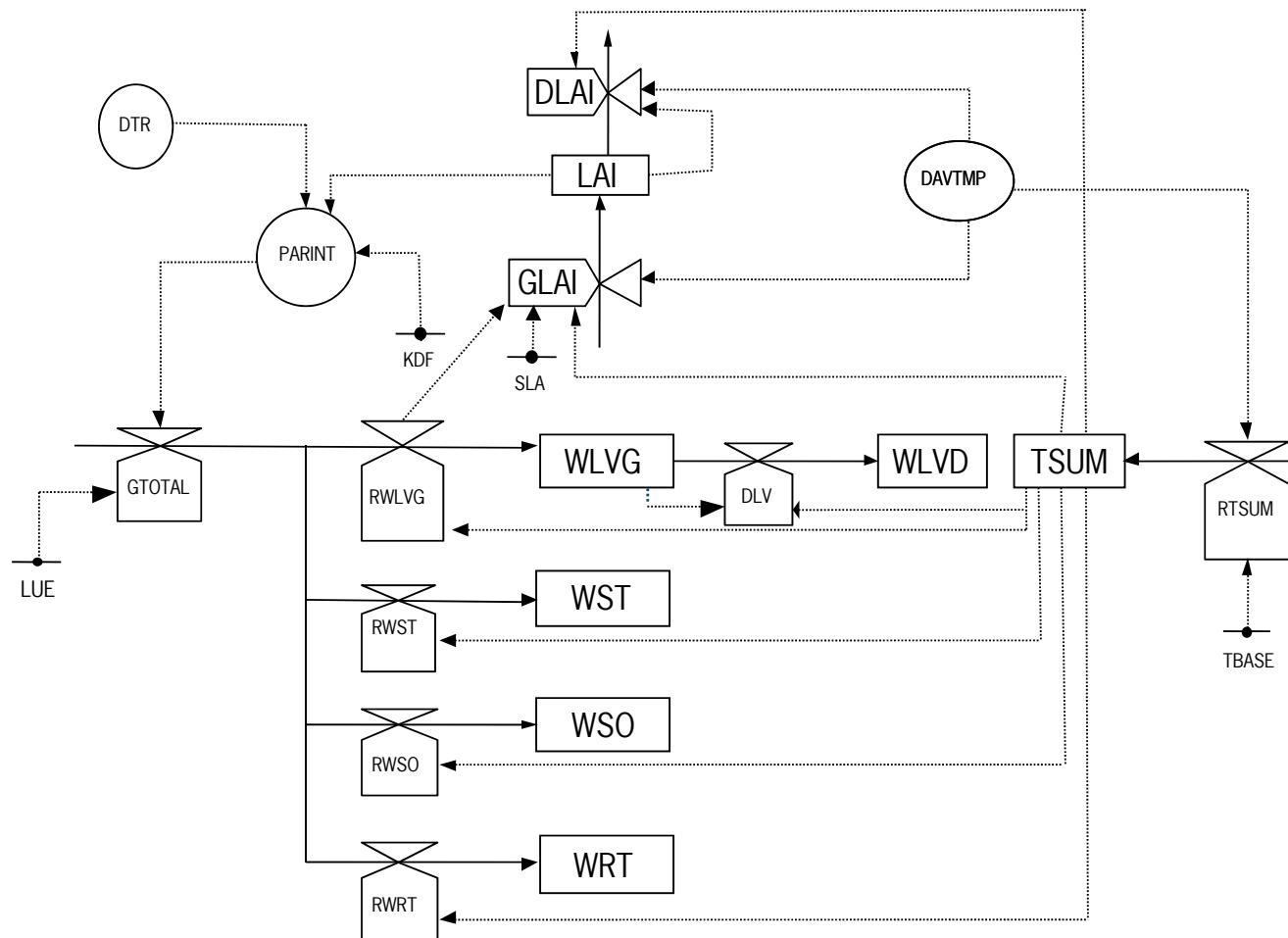


# Lintul1: Potential crop growth

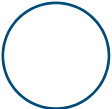
## Summarizing relational diagram and meaning symbols



# Relational diagram of Lintul1



# Forrester notation . . .

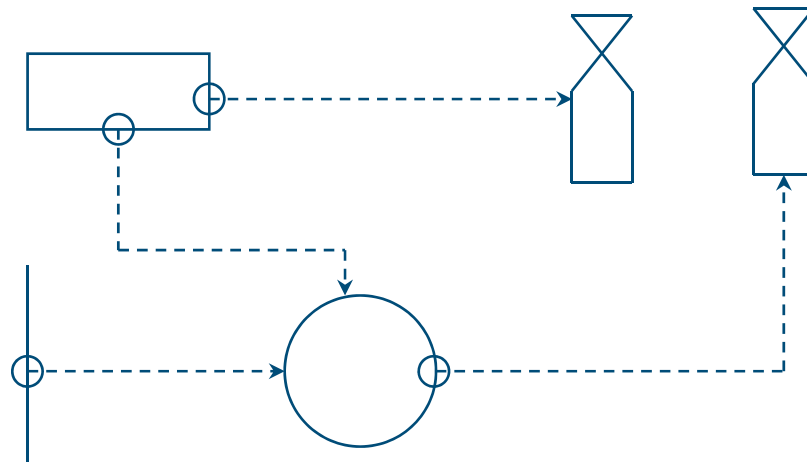
| Symbol                                                                              |                                                | Dimension            |
|-------------------------------------------------------------------------------------|------------------------------------------------|----------------------|
|    | Integral                                       | Amount               |
|    | Differential equation<br>(decision about rate) | Amount/time          |
|    | Flow of substance                              | Amount/time          |
|    | Discontinuous flow of<br>substance             | Amount/time          |
|    | Flow of information                            | No unit              |
|  | Input variable                                 | Not known beforehand |
|  | Auxiliary variable                             | Not known beforehand |

## . . . Forrester notation (continued)

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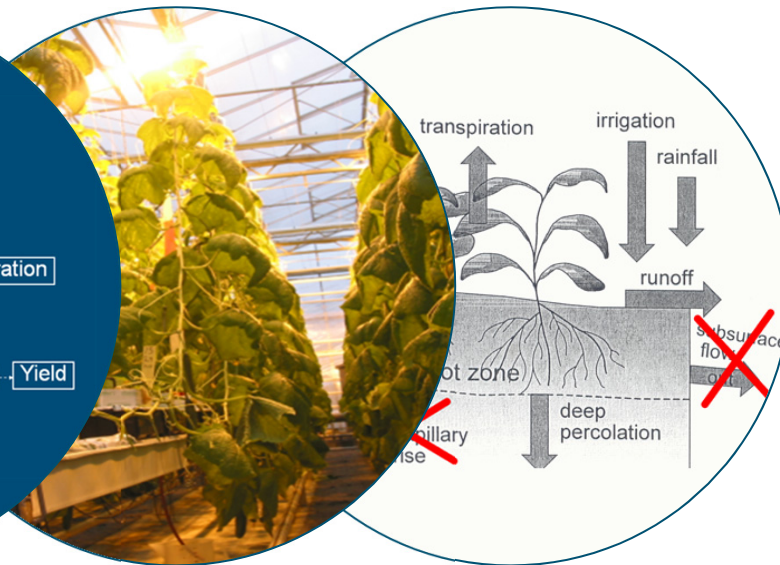
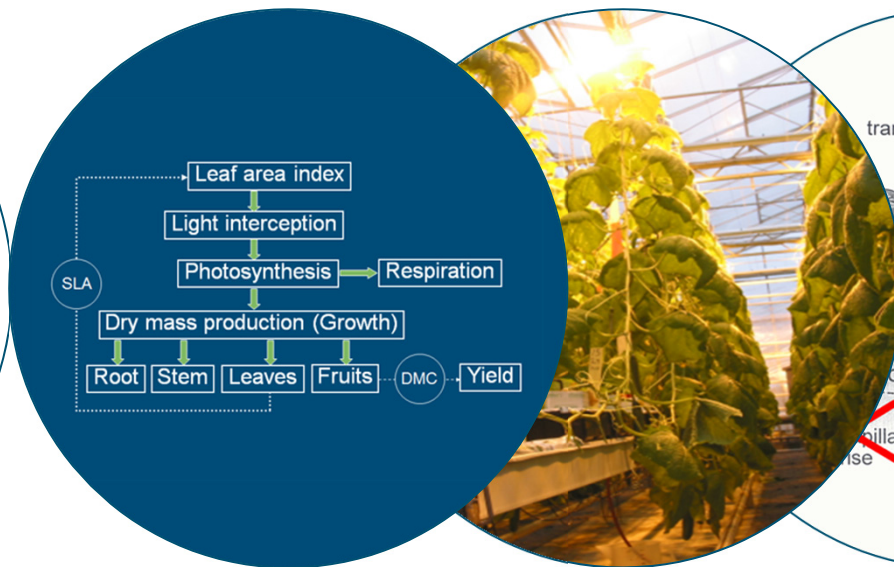
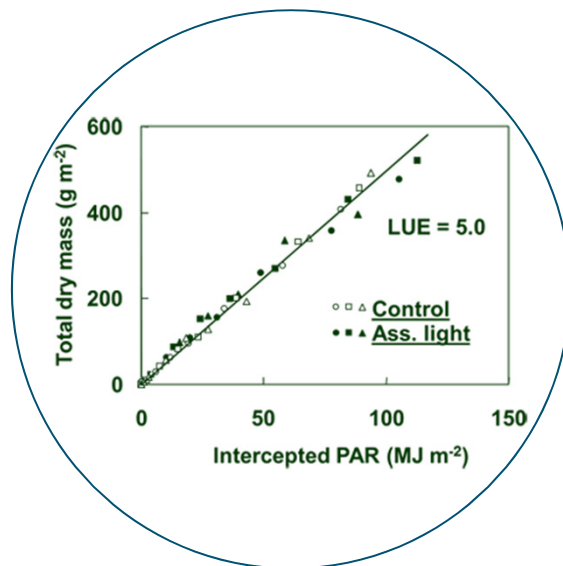
Information flows can only be used:

Information flows should point towards a place of decision, never to a state:



# Lintul1: Potential crop growth

## Exercises



# Practical exercises Lintul1

---

Exercises on the following subjects:

- Consequence of weather and different emergence dates on crop growth
- The sensitivity of yield for some important crop parameters or, differently stated, the consequence of uncertainties in parameter values
- Effects of parameter changes in light extinction (light capture), and light use efficiency on growth
- Potential yield as a function of emergence dates

There are data provided for a dry, an average and a wet year

Choose between Brazil and the Netherlands

Prepare a report for yourself on the PC to prepare for the exam



## To be able to do the exercises, study:

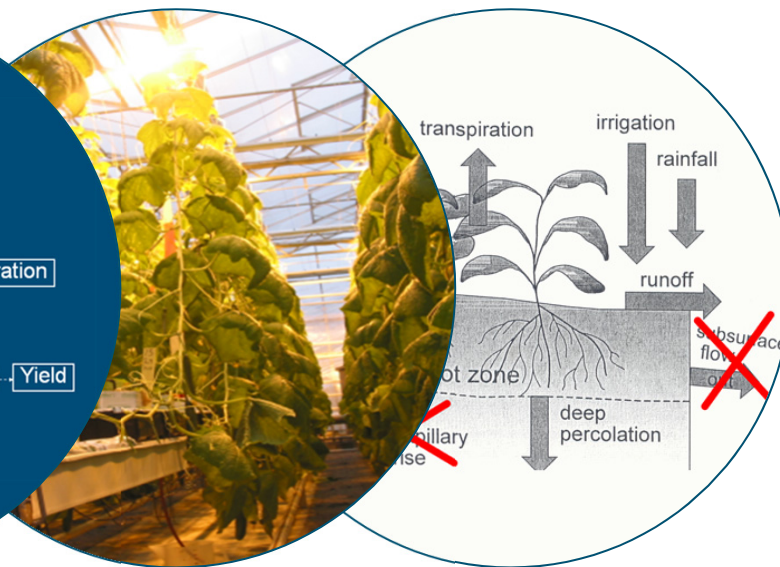
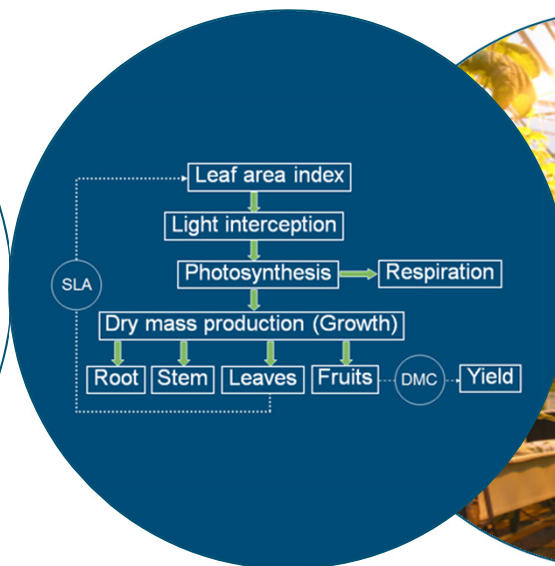
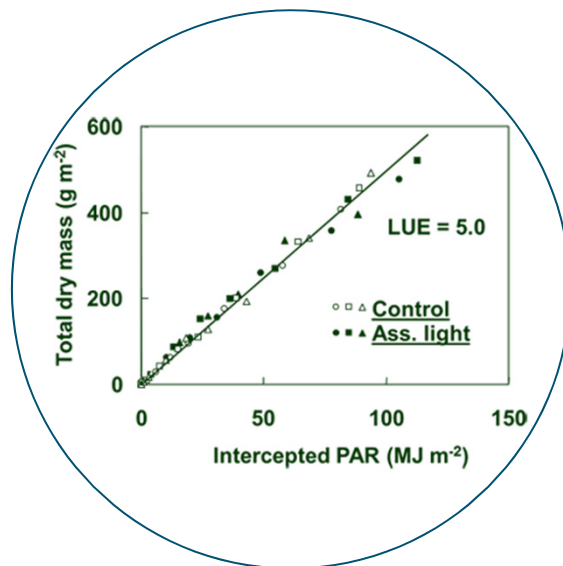
---

- Theory: "Lintul-1: potential crop growth"
- This power point presentation



# Lintul1: Potential crop growth

## References



# References used in the slides

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- Van Keulen, H. & N.G. Seligman, 1987. Simulation of water use, nitrogen nutrition and growth of a spring wheat crop. *Simulation Monographs*, Pudoc, Wageningen, pp. 310.
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